

SAP CPI Project 3 — XML Processing, XPath & Dynamic Expressions

Project Type: Core CPI Message Processing Flow

Status: Successfully Executed

Project Objective

This integration flow demonstrates how SAP CPI processes XML payloads internally using XPath extraction, Exchange Properties, Expressions, and Groovy logging.

Integration Flow

Timer Start → Create XML → Extract Properties → Generate Dynamic Output → Groovy Logging → End

Flow Explanation

Timer Start Event

The flow automatically starts using a Timer Start Event. A Timer was used because SAP CPI Trial environments may restrict inbound HTTP sender access.

Create XML Payload

A Content Modifier creates a sample XML order payload simulating enterprise order data.

XPath Property Extraction

XPath expressions extract Customer Name, Country, and Order Status values from the XML payload and store them as Exchange Properties.

Dynamic Expression Generation

A second Content Modifier dynamically generates a runtime message using the extracted properties and current system date.

Groovy Logging

A Groovy script logs the final output into SAP CPI Monitoring Attachments for debugging and monitoring.

XML Payload Used

```
<Order>
  <Id>1001</Id>
  <Customer>Vyshnavi</Customer>
  <Country>India</Country>
  <Status>URGENT</Status>
</Order>
```

XPath Properties

Property	XPath	Type	Data Type
country	/Order/Country	XPath	java.lang.String
orderstatus	/Order/Status	XPath	java.lang.String
customername	/Order/Customer	XPath	java.lang.String

Expression Used

Customer \${property.customername} placed a \${property.orderstatus} order from \${property.country} on \${date:now:yyyy-MM-dd}.

Final Output

Customer Vyshnavi placed a URGENT order from India on 2026-05-14.

Important Concepts Learned

- XML Payload Processing
- XPath Extraction
- Exchange Properties
- Dynamic Expressions
- Runtime Placeholders
- Groovy Logging
- Monitoring Attachments
- Message Object Basics

Real Enterprise Scenario

In real enterprise integrations, the XML payload would normally come from an external sender system through adapters such as HTTP, SOAP, or SFTP. For learning purposes, the payload was hardcoded using a Content Modifier.

Project Outcome

This project establishes the foundation for advanced SAP CPI development topics including Routing, Exception Handling, Data Stores, Message Mapping, and enterprise integration scenarios.